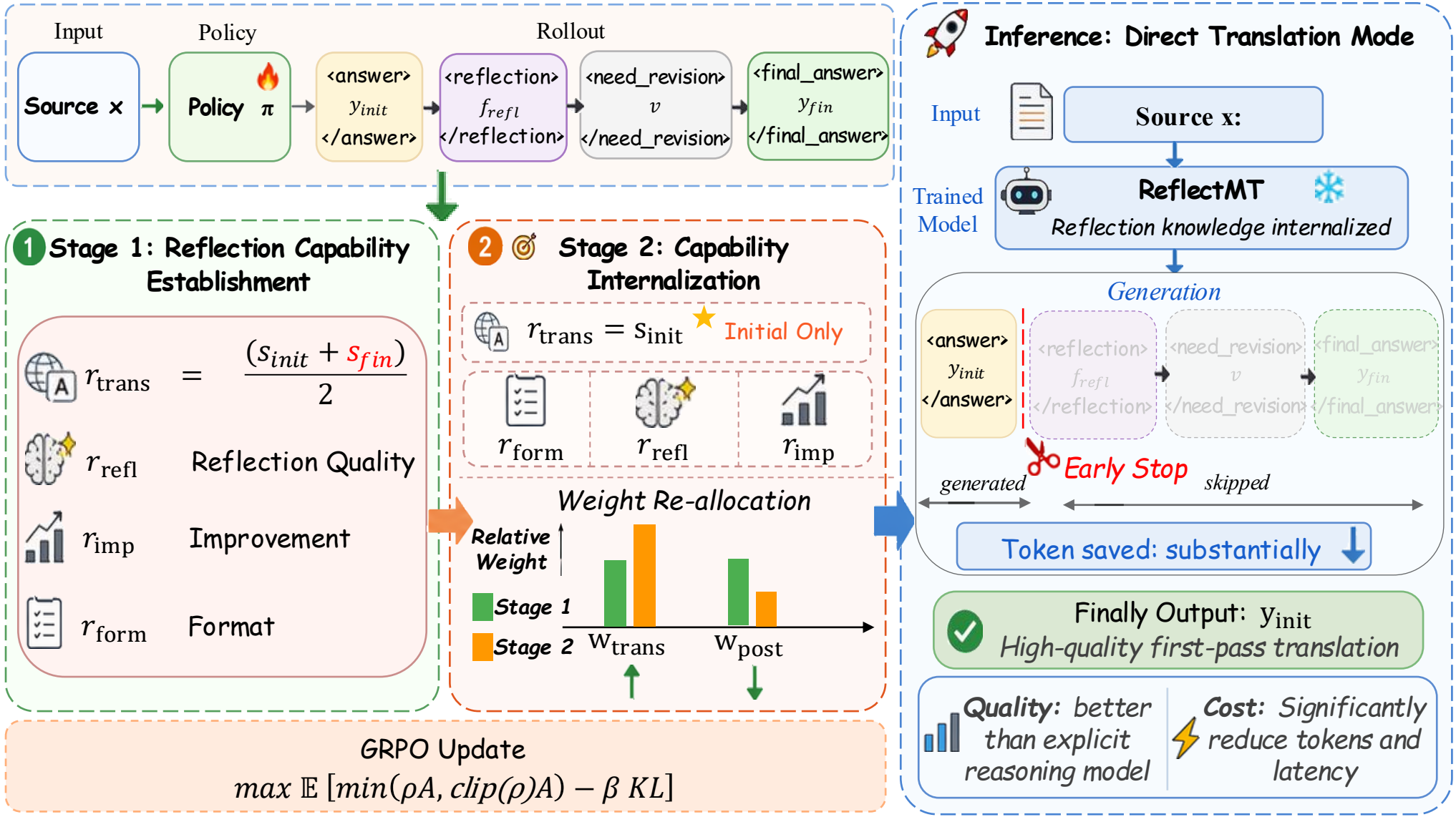

Input
Policy
Rollout
Source x
Policy π
<answer>
$y_{init}$
</answer>
<reflection>
$f_{refl}$
</reflection>
<need_revision>
$v$
</need_revision>
<final_answer>
$y_{fin}$
</final_answer>
1 Stage 1: Reflection Capability Establishment
$r_{trans} = \frac{(s_{init} + s_{fin})}{2}$
$r_{refl}$ Reflection Quality
$r_{imp}$ Improvement
$r_{form}$ Format
2 Stage 2: Capability Internalization
$r_{trans} = s_{init}$ Initial Only
$r_{form}$
$r_{refl}$
$r_{imp}$
Weight Re-allocation
Relative Weight
Stage 1
Stage 2
$w_{trans}$
$w_{post}$
GRPO Update
$max\ \mathbb{E}\ [min(\rho A, clip(\rho)A) - \beta\ KL]$
Inference: Direct Translation Mode
Input
Source x:
Trained Model
ReflectMT
Reflection knowledge internalized
Generation
<answer>
$y_{init}$
</answer>
<reflection>
$f_{refl}$
</reflection>
<need_revision>
$v$
</need_revision>
<final_answer>
$y_{fin}$
</final_answer>
Early Stop
generated
skipped
Token saved: substantially
Finally Output: $y_{init}$
High-quality first-pass translation
Quality: better than explicit reasoning model
Cost: Significantly reduce tokens and latency